Goal Programming

A new president has just been elected and has set the following economic goals (listed from highest to lowest priority):

- Goal 1: Balance the budget (this means revenues are at least as large as costs).
- Goal 2: Cut spending by at most $150 billion.
- Goal 3: Raise at most $550 billion in taxes from the upper class.
- Goal 4: Raise at most $350 billion in taxes from the lower class.

Currently, the government spends $1 trillion per year. Revenue can be raised in two ways: through a gas tax and through an income tax. You must determine G, the per-gallon tax rate (in cents); T_1 , the tax rate charged on the first $30,000 of income; T_2 , the tax rate charged on any income earned over $30,000; and C, the cut in spending (in billions). If the government chooses G, T_1 , and T_2 , then we assume that the revenue given in the table 3-1 below (in billions of dollars) is raised. Of course, the tax rate on income over $30,000 must be at least as large as the tax rate on the first $30,000 of income

	Low income	***High income***
Gas tax	G	0.5G
Tax on income up to $30,000	$20T_1$	$5T_1$
Tax on income over $30,000	0	$15T_2$

Table: Revenue raised for low income and high income

Problem statement:

The new president for the government wants to optimise the tax policy for economic growth. The president high priority is to avoid deficit i.e. cost should be less than revenue and reduce the spending by $150bn (from $1000bn). The revenue is generated through gas and income tax. We need to design tax system that satisfies the priorities set. This can be achieved using Goal programming technique.

Decision Variables

- G: Per-gallon gas tax rate (in cents)
- T1: Tax rate on the first $30,000 of income
- T2: Tax rate on income over $30,000
- C: Cut in spending (in billions)

***Deviation variables**:*

- d1+: Overachievement of balancing the budget
- d1-: Underachieving of balancing the budget
- d2+: Overachievement of cutting spending
- d3+: Overachievement of raising taxes from the upper class
- d4+: Overachievement of raising taxes from the lower class

Objective Functions:

Minimize $P1d1^{+} + P2d2^{+} + P3d3^{+} + P4d4^{+}$

Where P1 >> P2 >> P3 >> P4 are preemptive priorities.

Priority 1: Balance the Budget

Revenue >= Spending

$$G (20.5) + T1(25) + T2(15) >= 1000 – C$$

Priority 2: Cut Spending

$$C <= 150$$

Priority 3: Raise Taxes from Upper Class

$$0.5G + 5T1 + 15T2 <= 550$$

Priority 4: Raise Taxes from Lower Class

$$20G + 20T1 <= 350$$

Tax Rate Constraint:

$$T2 >= T1$$

Non-negativity:

$$G, T1, T2, C, d1+, d1-, d2-, d3-, d4- >= 0$$

(a) **Use preemptive goal programming to help the president meet their goals**

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Final Solution
C = 150.0
G = 300.0
T1 = 0.0
T2 = 26.666667
d1_neg = 0.0
d2_pos = 0.0
d3_pos = 0.0
d4_pos = 0.0
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The total revenue generated through Gas tax and income tax is **$850bn** after $150bn spending cut. The gas tax will be set at **300** cents or **$3/gallon**, and **26.67%** income tax will be imposed for the individuals earning over $30,000. There will be no tax imposed on the first $30,000 income, since all the set goals are achieved by the revenue generated from gas tax and tax from high income individuals.

(b) **Conduct a sensitivity analysis to determine how sensitive the revenue is to changes in G, T_1 , and T_2 . Which variable has the most significant impact on total revenue?**

We take the optimal results gained and adjust the variable by 1 percent change to analyze the variable that is most sensitive to the revenue.

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Sensitivity Analysis Results:
G (Gas Tax): Change in Revenue = 4.50 billion
T1 (Tax Rate on $30k): Change in Revenue = 0.00 billion
T2 (Tax Rate above $30k): Change in Revenue = 4.00 billion

Variable with the most significant impact: G (Gas Tax)
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The sensitivity analysis shows that the gas tax (G) has the most significant impact on total revenue, with a 1% increase generating **$4.50 billion**. The tax rate on income above $30,000 (T2) also significantly affects revenue, adding $4.00 billion for a 1% increase. In contrast, the tax rate on the first $30,000 (T1) has no impact due to its baseline value of zero. This highlights Gas tax as the most effective tool for revenue generation, followed by over $30k income tax, while low-income tax remains irrelevant in its current state.

(c) **Identify alternative techniques for handling the multi-objective optimisation problem. Compare and contrast these alternatives with the preemptive goal programming in terms of their underlying principles and methods of achieving multiple objectives**

We can use weighted sum method as an alternative to handle multi objective optimisation problem.

Weighted sum method

Principle:

The Weighted Sum Method would aggregate the different economic targets (budget balance, spending cuts, tax from upper and lower classes) under one overall objective function. Each goal would be assigned a weight based on relative importance, and the weighted objectives would be summed to form a final objective.

Methodology:

- In this case, the government would define weights for each goal, such as balancing the budget, cutting spending, and raising taxes from different classes. These weights would then be used to create a weighted sum of the other goals (e.g., balancing the budget would be multiplied by its weight, spending cut by its weight, etc.).
- The optimization would then seek to find the values for the gas tax rate (G), income tax rates (T1, T2), and spending cut (C) that maximize or minimize the weighted sum of these objectives.

Handling of Objective Importance:

- The assigned weights would reflect the importance of each economic goal (e.g., budget balance vs. tax revenues). For instance, balancing the budget might be given a higher weight than cutting spending, while revenue from the upper class could be weighted differently from the lower class.
- The method treats all objectives in a manner that indicates their relative importance through these weights, potentially balancing the trade-offs between them.

Goal programming

Principle:

Goal programming prioritizes the goals based on their importance. First is the balancing of the budget, followed by the second most important goal-such as cutting spending or raising taxes.

Methodology:

- Goal programming will focus primarily on achieving budget balance (Goal 1) first and any negative deviation from this goal will be minimized. Once the best result is achieved, the model will proceed to the next most important goal, such as limiting the spending cut (Goal 2), and so on.
- The decision variables—gas tax rate (G), income tax rates (T1, T2), and spending cuts (C)—would be adjusted in the order of priority, ensuring that the most important goals are satisfied before addressing the others.

Handling of Objective Importance:

- The Goal programming directly handles the importance of each goal. For example, the government's top priority is to balance the budget, so this goal would be solved first and must be satisfied as much as possible before considering the other goals.
- Goals like limiting spending cuts and raising taxes from the upper and lower classes would be addressed sequentially, with each lower-priority goal only considered after the higher-priority ones are satisfied to the greatest extent possible. This ensures that the most critical economic goals are achieved first, even if it means compromising on the lower-priority goals.